

Unveiling Key Clinical Indicators in Heart Disease: A Comprehensive Risk Assessment Framework

Abstract

This study uses advanced data visualization and statistical analysis methods to investigate the relationship between key clinical factors and heart disease. A comprehensive dataset, sourced from multiple heart disease databases, is analyzed to assess the predictive power of variables such as age, sex, cholesterol levels, blood pressure, and ECG readings in the context of cardiovascular risk. The study leverages Python's data science libraries, Matplotlib and Seaborn, to generate bar charts, heat maps, and combined factor analyses, providing new insights into how these variables contribute to heart disease. The study identifies high-risk patient profiles by integrating these visualizations with statistical models and enhances early detection capabilities. This research addresses a critical gap in previous studies by quantifying the impact of individual factors through precise percentage breakdowns and proposes a framework for improving clinical decision-making. The findings hold significant potential for reducing healthcare costs and improving patient outcomes through timely intervention.

Keywords: Heart disease, predictive modeling, data visualization, clinical risk factors, Python, statistical analysis, early detection

Introduction:

Heart Disease covers a wide scope of heart-related situations, including coronary artery disease (CHD), cardiac arrhythmias, and congestive heart failure (CHF)[1]. It is the primary reason for the worldwide fatality rate, reliable for roughly 17.9 million mortalities yearly, which comprises almost 33% of all worldwide mortalities[2]. The progress of heart disease is affected by several elements like heredity, hypertension, diabetic disorder, and daily routine traits[3].

Heart disease proceeds to implement a significant load on worldwide health services and financial systems [4]. The disorder not only influences ailment and fatality but continues to significantly burden worldwide health services and financial systems [4]. The disorder not only influences ailment and fatality but also leads to reduced quality and guides to reduced attributes of life and efficiency[5]. As cardiovascular disease improves, it regularly affects more complicated and costly therapies, emphasizing the requirement for early identification and protection attempts[6].

Early identification of heart disease is crucial for protecting against dangerous problems like ischemic heart disease and hemorrhagic stroke[7]. Early detection permits prompt clinical measures and daily routine alterations that can substantially decrease ailment and fatality[8]. Moreover, early identification and cure minimize healthcare expenses by blocking the requirement for more intrusive processes[9].

Forecasting heart disease permits healthcare suppliers to categorize patients by their threat stages, permitting focused procedures[10]. Machine learning frameworks and artificial intelligence types of equipment are being progressively utilized to forecast heart disease, displaying a commitment to enhancing analytical reliability [11]. Predictive frameworks can diminish the prevalence of extreme circulatory incidents, eventually recovering existence and minimizing healthcare expenses[3].

The current research concentrates on an extent of components that are broadly established as projected of heart disorders: Sex, age, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG, maximum heart rate, exercise-induced angina, oldpeak(ST depression), and ST slope [12]. These medical elements have been broadly researched for their connection with heart disease and perform a crucial role in threat evaluation structures[13].

This research aims to investigate the connection between the healthcare factors classified above and the occurrence of heart disease [5] Through statistical overview frameworks, this research aims to detect the most projective elements and improve early identification frameworks for heart disease[11].

Literature Review

The literature on heart disease risk prediction highlights the critical role of various clinical and demographic factors in early detection and patient stratification. Factors like Sex, Age, Chest pain Type, Resting Blood Pressure, Cholesterol, Fasting Blood Sugar, Resting ECG, Max Heart Rate, Exercise-Induced Angina, Oldpeak(ST Depression), and ST Slope contribute to early detection of heart disease and finding the most crucial factors.

Age is a significant component in cardiovascular threat, with males exhibiting significantly increased prevalence at younger years as compared to females. Males below the age of 55 are substantially more likely to encounter cardiovascular disorders as compared to females in the identical age set[1]. Sex disparities in heart disorder threat are strongly recorded, with males usually at greater risk at an earlier stage, while females encounter intensified threat at the postmenopausal stage[13]. Research has revealed that typical angina chest Pain is a crucial pointer of coronary heart disease and is correlated with increased fatality rate[14]. Hypertension and high blood pressure have consistently been detected as prominent threats to the progress of cardiovascular disease, as they promote the strengthening of capillaries[3]. Raised degrees of low-density lipoprotein (LDL) cholesterol are proximately connected to the composition of arterial plaques, which block circulation and enhance heart disease threat[9]. Diabetes and damaged glucose acceptance are robust indicators of cardiovascular disease because of their connection with circulatory immune response and injury[8]. irregular electrocardiogram (ECG) observations, like ST-segment anomalies, are estimates of hidden coronary artery disease and are regularly utilized in analytical specifications[11]. A decrease in maximum heart rate throughout the workout has been associated with poor heart health and an elevated threat of deleterious acute coronary syndromes (ACS)[7]. Angina accomplished throughout the fitness routine is implicative of coronary heart disease and is a robust indicator of future cardiac incidents[14]. The extent of ST depression (oldpeak) during anxiety inspection is utilized to evaluate cardiac ischemia and the threat of cardiovascular disease[12]. The ST Slope of the ST segment in an electrocardiogram is a vital marker of cardiac ischemia, with a flat or downsloping ST segment implying an increased probability of hypoxia[10].

Methodology:

The database utilized in this research was obtained from [Kaggle](#), particularly from the “Heart Disease Dataset” data collection. This Database was chosen by merging five distinct data collections from the following resources:

Table 1: Overview of Heart Disease Databases Used in the Study

Database	Instance
Cleveland Heart Disease	303
Hungarian Heart Disease Database	294
Switzerland Heart Disease Database	123
Long Beach VA Heart Disease Database	200
Stalog (Heart) Dataset	270

This collection outcomes in a total of 1190 instances, and 11 common attributes, crafting it into one of the biggest and most comprehensive databases accessible for cardiovascular disease research.

The database contains age, sex, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, resting ECG, maximum heart rate, exercise-induced angina, old peak (ST Depression), and ST slope. Each trait has been broadly investigated for its accord with heart disorder, making it appropriate for an analytical framework[15][16].

Table 2.Distribution of Key Clinical Factors in the Heart Disease Dataset

SR. NO.	Factors	Description	value
1.	Sex	Male	909
		Female	281
2.	Age	Young (≤ 40 years)	108
		Middle-aged (41–60 years)	789
		Senior (≥ 61 years)	293
3.	Chest Pain Type	Typical Angina	131
		Atypical Angina	165
		Non-Anginal Pain	269
		Asymptomatic	625
4.	Resting Blood Pressure	Normal	264
		Elevated	301
		High (HyperTension)	416
		Severe HyperTension	209
5.	Cholesterol	Desirable (Normal)	415
		Borderline High	265
		High	510

6.	Fasting Blood Sugar	Normal	936
		Elevated	254
7.	Resting ECG	Normal	684
		Abnormal	262
		Hypertrophy	244
8.	Max Heart Rate	Low(<130 bpm)	188
		Normal(130-160 bpm)	838
		High(>160 bpm)	164
9.	Exercise-Induced Angina	Yes	461
		No	729
10.	Oldpeak (ST Depression)	No Depression	455
		Mild Depression (0.5-1.5)	278
		Moderate Depression (1.6-2.5)	205
		Severe Depression (>2.5)	252
11.	ST Slope	Upsloping	365
		Flat	582
		Downsloping	243
12.	Heart Disease	Disease	629
		No Disease	561

These traits form the base for heart disease threat assessment analysis [17].

The database utilized in this research is extremely suitable for the research objective, as it encompasses globally acknowledged medical elements like age, sex, cholesterol level, and blood pressure, all of which are indicators of cardiovascular disease. These traits, illustrated by an extensive set of cardiovascular disease databases, offer a vigorous infrastructure for analysis, securing the outcomes suitable to real-world heart disease threat evaluation and early identification of framework[1] [3][15].

Results and Discussion

In this research, Python was utilized to analyze the data and create visualizations using libraries such as Matplotlib and Seaborn. These tools are authorized for the creation of bar charts and heat maps to

efficiently represent the relationships between diagnostic factors and heart disease. The graphical representations give clear perspectives into the distribution of threat factors, facilitating a more comprehensive understanding of their effect."

1.Age

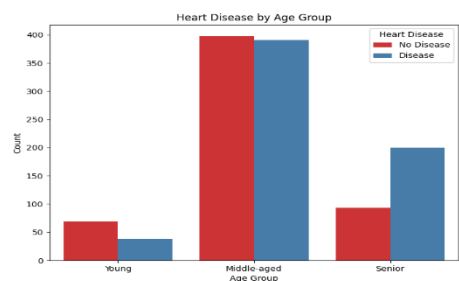


Fig 1.Age Distribution and Heart Disease Risk by Age Group

Fig 1 bar chart illustrates that heart disease is the most dominant in the middle-aged (41-60) years. The threat grows especially from the younger group (≤ 40 years) to the senior group (≥ 61 years), where cardiovascular disease becomes more typical due to age correlated with cardiac decrease.

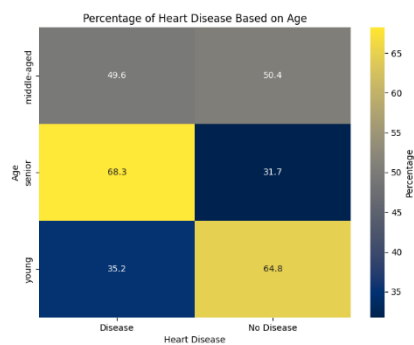


Fig 2.Heatmap of Heart Disease Risk Concentration by Age Group

Fig 2 heat map illustrates that 49.6% of heart disease risk cases are condensed in the middle-aged group, while 68.3% of cases are discovered in the senior group condensed as very high-risk. The young group has the minimum rate, auditing only 35.2% of cases, ruminating the lower risk in younger entities.

2. sex

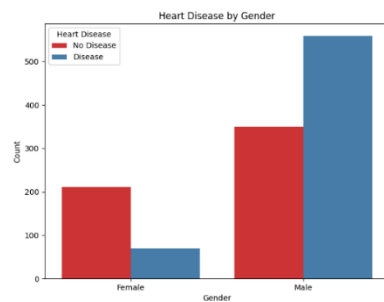


Fig 3. Gender Distribution and Heart Disease Prevalence

Fig 3 illustrates that males have a substantially higher extent of cardiac disease as compared to females, particularly in the middle age group. Postmenopausal females begin to grab males in terms of cardiac disease, but comprehensive males continue at a higher threat.

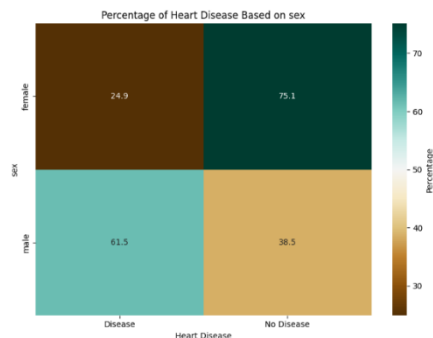


Fig 4. Contribution of Males and Females to Overall Heart Disease Cases

Fig 4. Demonstrates that men account for 61.5% of the overall cardiovascular disease cases, with the topmost intentness in middle-aged and senior men. Women contribute only 24.9% with the threat increasing after menopause.

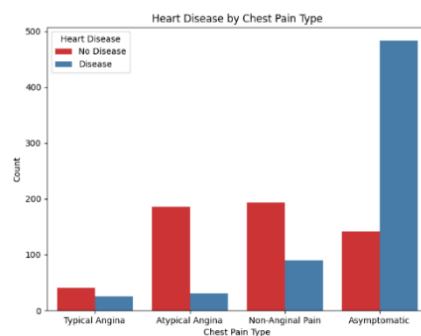


Fig 5. Chest Pain Type and Its Predictive Power for Heart Disease

Fig 5. Demonstrates that in Asymptomatic pain is the most prognostic symptom of heart disease with a significant risk than Non-anginal pain and Atypical pain shows more prevalence in heart

disease in terms of cases after that typical angina also plays a vital role in making the risk of cardiac disease.

Fig 6. Distribution of Heart Disease Cases by Asymptomatic and Symptomatic Pain

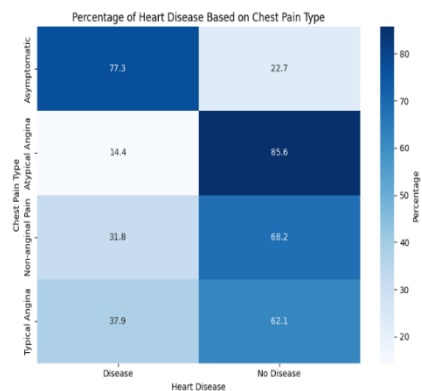


Fig 6 demonstrates that asymptomatic pain makes up 77.3 % of cases of heart disease, highlighting the higher risk even when symptoms are not present after 37.9 % risk shown in typical angina also shows a prevalence in heart disease

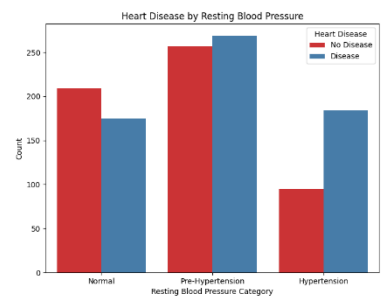


Fig 7.Relationship Between Resting Blood Pressure and Heart Disease

Patients with higher resting blood pressure (≥ 140 mmHg)/prehypertension have more potential to be afflicted with cardiac disease compared to those with normal blood pressure. Hypertension, a key threat factor, arises regularly amidst heart disease patients.

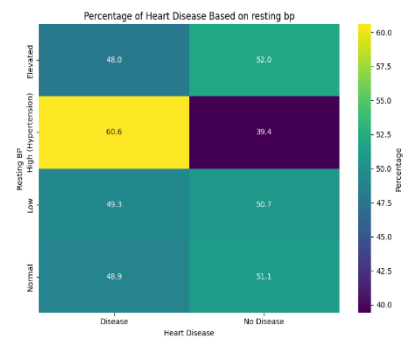


Fig 8. □ Prevalence of Heart Disease Cases by Resting Blood Pressure Level

60.6% of heart disease cases occur in individuals with high resting blood pressure. Only 48.9 % of cases are discovered in individuals with normal blood pressure levels, authenticating hypertension task as an important threat factor where low and elevated shows 49.3 % and 48.0% incident to heart disease.

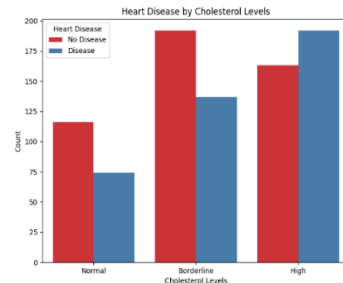


Fig 9.Heart Disease Risk Based on Cholesterol Levels

Individuals with high cholesterol levels (≥ 240 mg/dL) are more prevalence to have heart disease. While cholesterol is a key element, it seems less independently statistical as compared to other factors like chest pain and maximum heart rate.

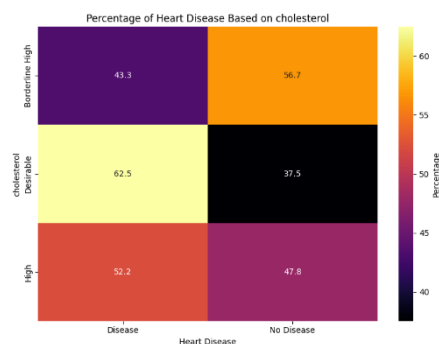


Fig 10. Cholesterol Levels and Their Association with Heart Disease Risk

62.5% of heart disease cases are connected with cholesterol levels < 200 mg/dl that stand as very high risk whereas 52.5% of cases occur in individuals with higher cholesterol levels, and 43.3% of cases appear in borderline high cholesterol levels showing the threat of cardiac disease.

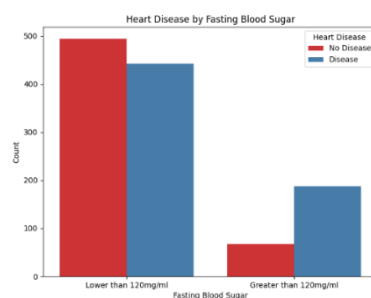


Fig 11. Fasting Blood Sugar Levels and Heart Disease Risk

fasting blood sugar levels (≥ 120 mg/dL) are strongly associated with heart disease, especially in diabetic patients. The higher the fasting blood sugar, the greater the probability of cardiac disease.

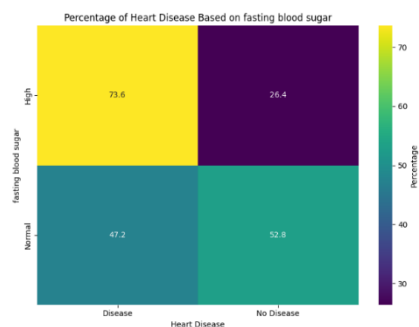


Fig 12. Prevalence of Heart Disease by Fasting Blood Sugar Levels

73.6% of heart disease cases appear in patients with high fasting blood sugar levels, strengthening the connection between diabetes and cardiovascular disease. This emphasizes the require for glucose management to decrease heart disease threat.

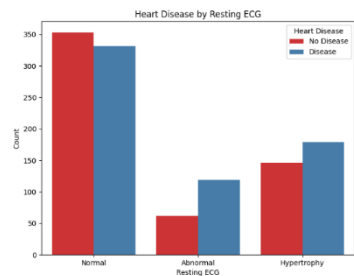


Fig 13. ECG Readings and Their Impact on Cardiovascular Disease Identification

Normal ECG readings show more prevalence of cardiovascular disease cases after hypertrophy and abnormal ECG shows key values in identifying the cardiovascular disease identification.

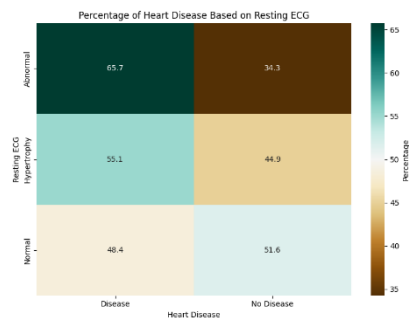


Fig 14. Risk of Heart Disease in Patients with Abnormal ECG

Abnormal ECG (ST-segment changes or other abnormalities) with 65.7% is usual among heart disease patients. With higher risk. The presence of the irregularities shows underlying hypoxia whereas 55.1% seen in hypertrophy ECG indicates a vital role in threat measurement of cardiac disease.

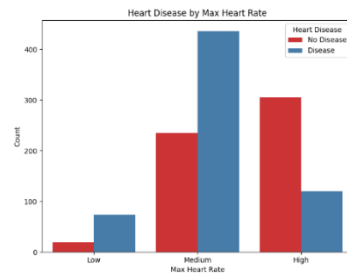


Fig 15 Maximum Heart Rate and Its Correlation with Heart Disease

A medium maximum heart rate (100-160 bpm) is highly linked with heart disease. Individuals with heart disease exhibit constantly lower heart rates, highlighting poorer cardiovascular health and bad fitness levels.

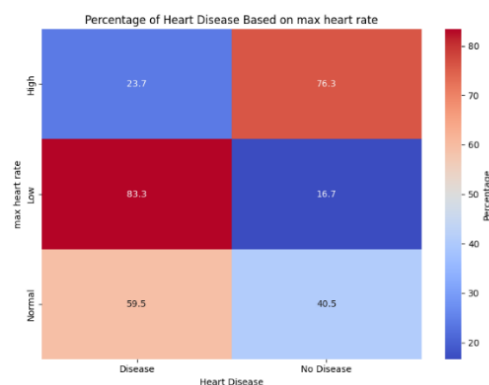


Fig 16. Prevalence of Heart Disease in Patients with Low Maximum Heart Rate

83.3 % of heart disease cases are discovered in patients with a maximum heart rate below 100 bpm, making it a crucial predictor of cardiovascular disease. Higher heart rates are minimally linked with heart disease, aiding the priority of heart conditions.

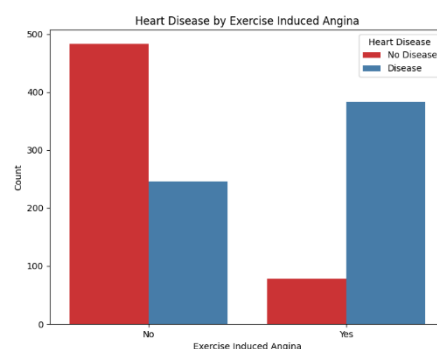


Fig 17. Exercise-Induced Angina as an Indicator of Heart Disease

Patients who encounter angina (chest pain) during physical exercise are more likely to have cardiovascular disease. Exercise-induced angina usually symptoms oxygen-deprived heart disease, where the heart doesn't get sufficient blood flow during exercise.

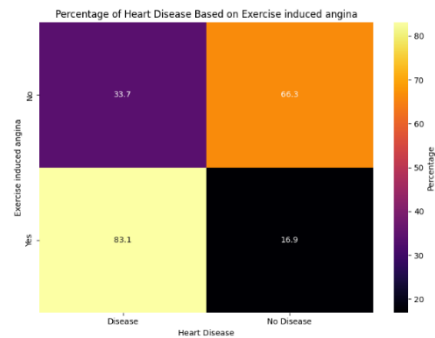


Fig 18. Prevalence of Exercise-Induced Angina in Heart Disease Patients

83.1% of cardiac disease patients encounter exercise-induced angina, making it a robust indicator of cardiac heart disease. This strengthens the value of exercise anxiety assessment in analyzing cardiac situations.

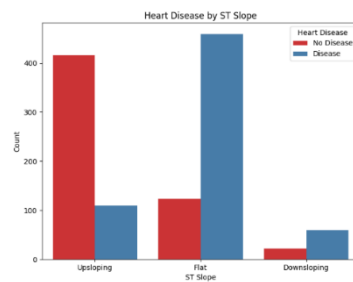


Fig 19. ST Segment Slope and Its Connection with Heart Disease Risk

A flat ST segment is vigorously connected with cardiovascular disease increasing the risk. Patients with a downsloping ST slope also play a vital role in suffering from hypoxic heart disease as compared to upsloping.

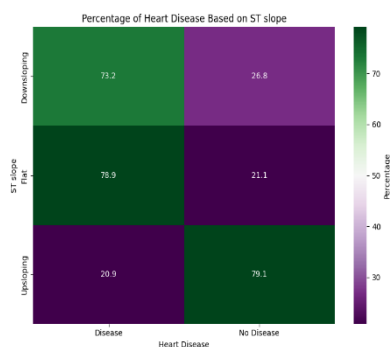


Fig 20. Prevalence of Heart Disease Based on ST Segment Patterns

78.9% of cardiovascular disease cases are detected in individuals with a flat ST segment, substantiating its task as a key analytical tool for detecting hypoxic heart situations. Downsloping with 73.2% also increasing the threat of cardiac disease.

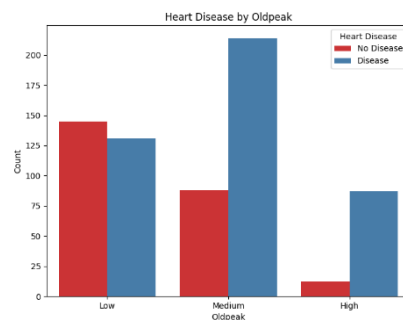


Fig 21.ST Depression Levels During Stress Test and Heart Disease Risk

Medium level of ST depression during a stress test is strictly connected to heart disease. Higher levels of ST depression are associated with more acute coronary artery hindrances.

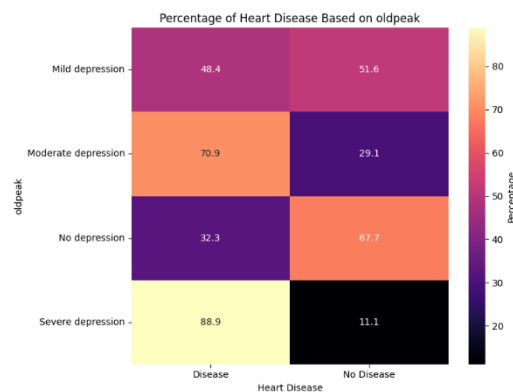


Fig 22.Prevalence of Heart Disease by Severity of ST Depression

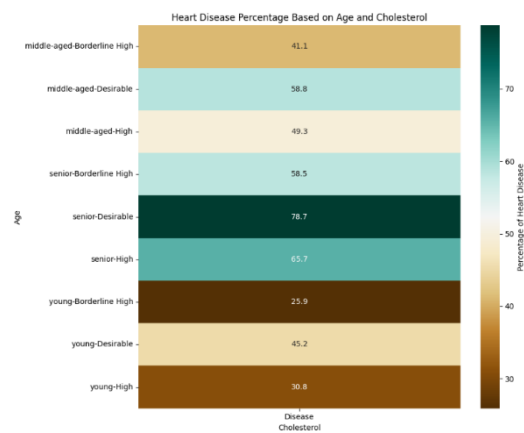
88.9% of heart disease cases entail an Oldpeak of severe depression, showing influential ischemia and very high risk. This strengthens the efficiency of stress testing in examining the intensity of heart disease. Moderate depression with 70.9% boosts the risk of cardiac disorder as compared to mild depression.

Critical Section

In our further research, we combine multiple factors because heart disease is multifactorial. By combining two factors they identify a more comprehensive investigation of cardiovascular

health. The combined impact of selected factors provides better examination accuracy and more focused risk identification.

Fig 23. Combined Impact of Age and Cholesterol on Heart Disease Risk



This graph demonstrates how increasing age and higher cholesterol levels are intensely connected with heart disease with 78.7 and 58.5%. Heart disease occurrence is likely greatest among older people with elevated cholesterol (e.g., ≥ 240 mg/dL). Younger individuals with higher cholesterol levels show a diminished connection with heart disease. This combination indicates the aggregate risk associated with aging and cholesterol.

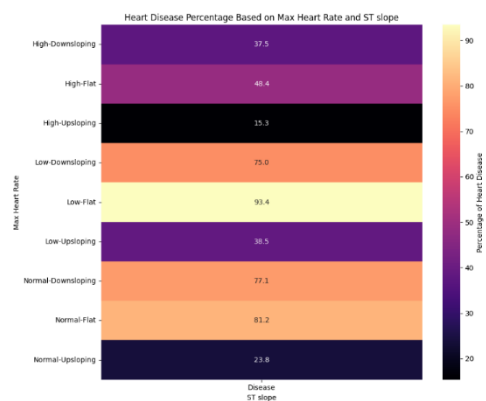


Fig 24. Low Maximum Heart Rate and Abnormal ST Slope in Heart Disease Prediction

This graph illustrates that lower maximum heart rates (e.g., <100 bpm) during endeavor and a flat or downsloping ST slope are connected with a greater percentage of cardiac disease cases. Individuals who demonstrate both a low maximum heart rate and an abnormal ST slope have a substantially greater risk, signifying reduced cardiac execution and ischemic stress. The alliance of these factors is critical because it analyzes both functional capacity and ischemic feedback.

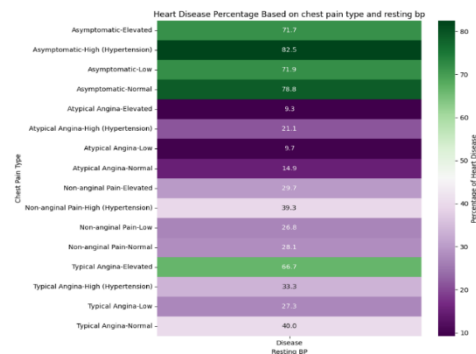


Fig 25. Resting Blood Pressure and Asymptomatic Pain in Heart Disease Prediction

The graph illustrates that asymptomatic pain with high resting blood pressure (e.g., ≥ 140 mmHg) and at normal blood pressure (120–129 mmHg) indicate a greater incidence of heart disease with 82.5% and 78.8%. In variance, those with typical angina chest pain and elevated blood pressure likely show high risk with 66.7%. This combination is critical because the chest pain type provides instant clinical symptoms of cardiac disease while resting blood pressure gives intuition into long-term cardiovascular health.

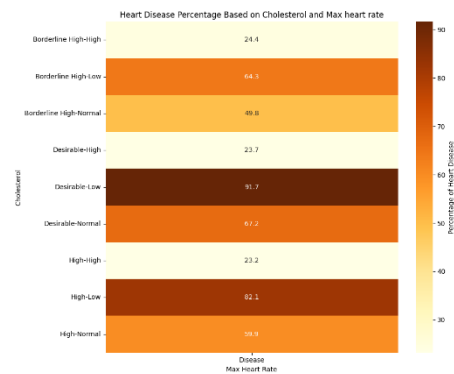


Fig 26. Combined Impact of Maximum Heart Rate and Cholesterol Levels on Heart Disease Risk

This graph demonstrates that individuals who had a low max heart rate with desirable cholesterol (< 200 mg/dl) show a higher risk of heart disease at 91.7% which is very crucial for health and a person with a low max heart rate and high cholesterol level (≥ 240 mg/dl) indicates a strong predictor of heart disease with 82.1%. Combining this two factor illustrates the higher threat and indicator of cardiac disease.

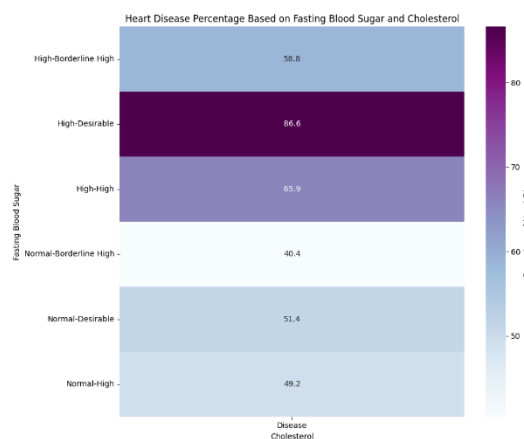


Fig 27. Fasting Blood Sugar and Cholesterol Levels and Their Combined Impact on Heart Disease Risk

This figure illustrates that individuals with high fasting blood sugar (e.g., ≥ 120 mg/dL) and desirable cholesterol (< 200 mg/dL) have a greater risk of heart disease at 66.6%. also, both high blood sugar and high cholesterol levels show greater risk with 65.9%. This combination points to a significant high risk among diabetic patients with uncontrolled lipid levels. The combination of high blood sugar and cholesterol is particularly dangerous because both conditions correlate to blood vascular damage and atherosclerosis.

Conclusion

In conclusion, this study highlights the importance of early detection in managing heart disease by identifying key risk factors such as age, sex, cholesterol levels, and ECG readings. The findings demonstrate that middle-aged and older individuals, especially men, are at higher risk. Predictive models using these factors can enhance diagnostic accuracy, allowing for timely interventions that reduce both mortality and healthcare costs. Continued refinement of these models will further improve early detection and patient outcomes.

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